5.2 Few-shot learning

Having explored the various types of prompts and techniques for testing them, we can now focus on practical prompt engineering techniques that you can implement without the need for fine-tuning or configuring the LLM models you interface with. These methods will allow you to optimize AI outputs efficiently and effectively, enhancing the performance of your generative AI applications.

We start with few-shot learning: this is a simple yet powerful technique in prompt engineering that allows AI models to generalize and adapt to new tasks with just a few examples, rather than requiring extensive training on large datasets.

Imagine trying to teach someone how to cook a specific recipe. Instead of providing them with a full cookbook or a lengthy set of instructions, you simply give them a few examples of how similar dishes are prepared. For instance, if you want them to make a pasta dish, you might show them how to cook spaghetti carbonara and arrabbiata. Depending on the relevance of these examples, they can infer the techniques, ingredients, and general approach needed to create the dish you want.

Zero-shot vs. few-shot vs. fine-tuning

In addition to few-shot learning, there are several other learning techniques used in generative AI, including zero-shot learning and fine-tuning. Each of these methods has its own unique approach and use cases, which can significantly affect the performance of AI models. The following is a brief overview of these techniques and how they differ from few-shot learning:

- *Zero-shot learning* is a technique where the model is tasked with performing a task without seeing specific examples related to that task beforehand. Instead, it relies on its preexisting knowledge and understanding of language to generate responses.
- *Fine-tuning,* on the other hand, involves taking a pretrained model and further training it on a specific dataset to improve its performance on a particular task or domain. This process adjusts the model's parameters based on the new data. If the operation is successful, it often results in a more accurate model at the cost of being more resource intensive and requiring significant computational power and time.

Before I explain how to implement few-shot learning, let's visualize the general steps involved for crafting prompts using few-shot learning as an example. Based on figure 5.5, we can identify the following steps:

1. *User interaction*—We provide a few examples that demonstrate the desired output format or content.
2. *Provide examples*—We input examples that the model can reference. For instance, we might provide examples of summaries related to AI. This process would be the most crucial, since it will determine the outcome.